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$$4.25 (a) X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt = \int_{-1}^3 x(t) e^{-j\omega t} dt = \int_{-1}^0 2e^{-j\omega t} dt + \int_0^2 e^{-j\omega t} dt + \int_2^3 (2-t) e^{-j\omega t} dt + \int_3^{\infty} 0 dt$$

$$= -\frac{2}{j\omega} (e^{-j3\omega} - e^{-j2\omega} + e^{-j\omega} - e^{j\omega})$$

$$= -\frac{2}{j\omega} (e^{-j3\omega} - e^{-j2\omega} + e^{-j\omega} - e^{j\omega}) + (-\frac{1}{j\omega}) [te^{-j\omega t} \Big|_1^2 + te^{-j\omega t} \Big|_1^0 - (-\frac{1}{j\omega}) (e^{-j\omega t} \Big|_1^2 + e^{-j\omega t} \Big|_1^0)]$$

$$= -\frac{1}{j\omega} [2e^{-j3\omega} - 2e^{-j2\omega} + 2e^{-j\omega} - 2e^{j\omega} + 2e^{-j2\omega} - 2e^{-j\omega} + 2e^{-j\omega} - e^{-j2\omega} - e^{-j\omega} + \frac{1}{j\omega} (e^{-j2\omega} + 1 - 2e^{-j\omega})]$$

$$= -\frac{1}{j\omega} [2e^{-j3\omega} - 2e^{j\omega} + \frac{1}{j\omega} (e^{-j2\omega} - 2e^{-j\omega} + 1)]$$

$$= \frac{1}{j\omega} (2e^{j\omega} - 2e^{-j3\omega}) + \frac{1}{\omega^2} (e^{-j2\omega} - 2e^{-j\omega} + 1)$$

$$= \frac{2}{j\omega} ((\cos\omega + \cos 3\omega) + j(\sin\omega + \sin 3\omega)) + \frac{1}{\omega^2} (e^{-j2\omega} - 2e^{-j\omega} + 1)$$

$$= \frac{2\sin 2\omega}{j\omega} (\sin\omega + j\cos\omega) + \frac{1}{\omega^2} (2\cos^2\omega - 2\cos\omega - j(\sin 2\omega - 2\sin\omega))$$

$$= \frac{4\sin 2\omega}{\omega} (\cos\omega - j\sin\omega) + \frac{2}{\omega^2} (\cos\omega - 1) (\cos\omega - j\sin\omega)$$

$$= \left(\frac{4\sin 2\omega}{\omega} + \frac{2}{\omega^2} (\cos\omega - 1) \right) e^{-j\omega} = 4 \left(\frac{\sin 2\omega}{\omega} - \frac{\sin^2 \frac{\omega}{2}}{\omega^2} \right) e^{-j\omega}$$

$$X(j\omega) = \left\{ \frac{-\omega}{2\pi} \right\} \quad (\{x\} \text{ 是取小数部分}, \{x\} := x - [x])$$

$$(b) X(j0) = \lim_{\omega \rightarrow 0} 4 \left(\frac{\sin 2\omega}{\omega} - \frac{\sin^2 \frac{\omega}{2}}{\omega^2} \right) = 4 \times \left(2 - \frac{1}{4} \right) = 7$$

$$(c) X(0) = \int_{-\infty}^{+\infty} X(j\omega) d\omega = 2$$

$$(d) \text{ 记 } y(t) = \begin{cases} 1, & 1 \leq t \leq 3 \\ 0, & t < 1 \text{ 或 } t > 3 \end{cases} \quad Y(j\omega) = \frac{2\sin\omega}{\omega} e^{-j2\omega}$$

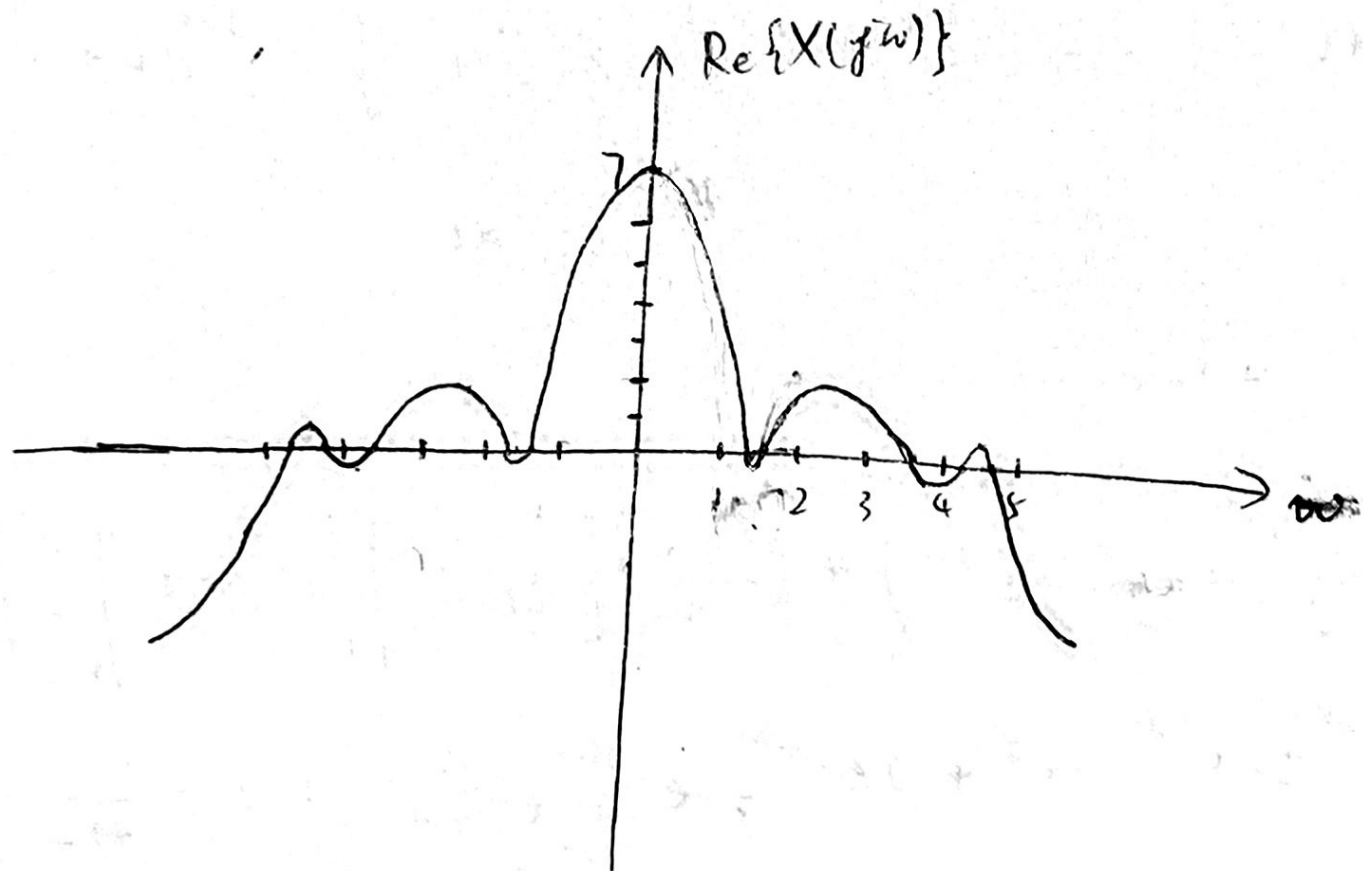
$$(e) R(j\omega) = \int_{-\infty}^{+\infty} x(t) y(t) e^{-j\omega t} dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) Y(\omega - \theta) d\theta = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\theta) \cdot \frac{2\sin(\omega - \theta)}{\omega - \theta} e^{-j2(\omega - \theta)} d\theta$$

$$\Rightarrow \int_{-\infty}^{+\infty} X(j\omega) \cdot \frac{2\sin\omega}{\omega} e^{j2\omega} d\omega = R(j0) \cdot 2\pi = 2\pi \int_{-\infty}^{+\infty} x(t) y(t) dt = \left(\frac{(1+2) \times 1}{2} + 1 \times 2 \right) \times 2\pi = 7\pi$$

$$(e) \int_{-\infty}^{+\infty} |X(j\omega)|^2 d\omega = 2\pi \int_{-\infty}^{+\infty} |x(t)|^2 dt = 2\pi \int_{-1}^3 |x(t)|^2 dt = 2\pi \left(4 \times 2 + 2 \times \int_1^2 t^2 dt \right)$$

$$= 4\pi \left(4 + \frac{8-1}{3} \right) = 4\pi \times \frac{19}{3} = \frac{76}{3} \pi$$

$$(f) \operatorname{Re}\{X(j\omega)\} = 4 \left(\frac{\sin 2\omega}{\omega} - \frac{\sin^2 \frac{\omega}{2}}{\omega^2} \right) \cos \omega$$





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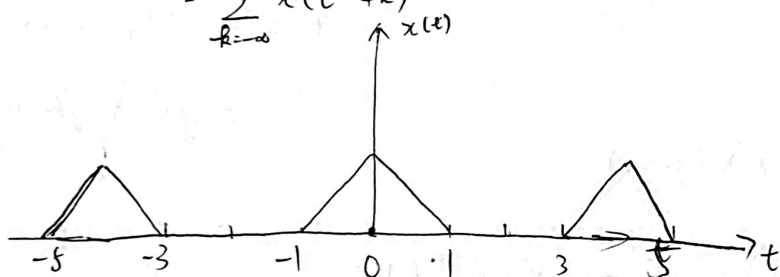
4.37 (a) $X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt = \int_{-1}^0 (t+1) e^{-j\omega t} dt + \int_0^1 (1-t) e^{-j\omega t} dt$

$$= \int_0^1 (1-t) e^{j\omega t} dt + \int_0^1 e^{-j\omega t} (1-t) dt = 2 \int_0^1 (1-t) \cos \omega t dt$$

$$= \frac{2}{\omega} \left((1-t) \sin \omega t \Big|_0^1 + \int_0^1 \sin \omega t dt \right) = \frac{2}{\omega^2} \cos \omega t \Big|_1^0 = \frac{2}{\omega^2} (1 - \cos \omega)$$

(b) $\tilde{x}(t) = \int_{-\infty}^{+\infty} x(\tau) \sum_{k=-\infty}^{+\infty} \delta(t-\tau-4k) d\tau = \sum_{k=-\infty}^{+\infty} \int_{-\infty}^{+\infty} x(\tau) \delta(t-4k-\tau) d\tau$

$$= \sum_{k=-\infty}^{+\infty} x(t-4k)$$



(c) $g(t) = x(t-4m), m \in \mathbb{Z}$

~~(d) $G(j\omega) = \int_{-\infty}^{+\infty} x(t-4m) e^{-j\omega t} dt = X(j\omega) e^{-j4\omega m} \neq X(j\omega)$~~

~~当 $4\omega m = 2k\pi$ 时, 即 $\omega = \frac{k\pi}{2m} = \left(\frac{k}{m}\right) \cdot \frac{\pi}{2}$~~

~~(d) $G(j\omega) = \int_{-\infty}^{+\infty} g(t) e^{-j\omega t} dt$~~

~~$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$~~

~~$\Rightarrow G(j\omega) - X(j\omega) = \int_{-\infty}^{+\infty} (g(t) - x(t)) e^{-j\omega t} dt$~~

~~$\int_{-\infty}^{+\infty} (x(t) - g(t)) \sum_{k=-\infty}^{+\infty} \delta(t-\tau-4k) d\tau = 0 = \sum_{k=-\infty}^{+\infty} [x(t-4k) - g(t-4k)]$~~

(d) 易知 $g(t) = x(t-4m), m \in \mathbb{Z}$

故 $G(j\omega) = X(j\omega) e^{-j4\omega m}$. 所以 $G(j\omega) = X(j\omega) \Leftrightarrow \omega = \frac{k\pi}{2m} = \left(\frac{k}{m}\right) \cdot \frac{\pi}{2}$

~~取 $k=m=1$ 时, $\omega = \frac{\pi}{2}$~~ , 取 $k=2m$, 则 $\omega = \frac{n\pi}{2}$.

$\Rightarrow X(j \cdot \frac{n\pi}{2}) = G(j \cdot \frac{n\pi}{2})$

$$4.39 \quad (a) \quad \cancel{G(j\omega)} \quad G(j\omega) = \int_{-\infty}^{+\infty} g(t) e^{-j\omega t} dt = \int_{-\infty}^{+\infty} x(jt) e^{-j\omega t} dt$$

$$\cancel{- \int_{-\infty}^{+\infty} x(-jt) e^{-j\omega t} dt = \int_{-\infty}^{+\infty} x(j(-t)) e^{j(-t)(-\omega)} dt}$$

$$= \int_{-\infty}^{+\infty} x(jt) e^{jt(-\omega)} dt = 2\pi x(-\omega)$$

$$(b) \quad \delta(t+B) \rightarrow e^{jB\omega}$$

$$\mathcal{F}(e^{jBt}) = 2\pi \delta(B-\omega) = 2\pi \delta(\omega-B)$$

$$4.40 \quad x(t) \rightsquigarrow X(j\omega) = \tilde{X}(j\omega, n)$$

$$\begin{aligned} \frac{dx(t)}{dt} &= \frac{t^{n-2}}{(n-2)!} e^{-at} u(t) + (-a) x(t) + \frac{t^{n-1}}{(n-1)!} e^{-at} \delta(t) \rightsquigarrow \tilde{X}(j\omega, n-1) - a\tilde{X}(j\omega, n) \\ &+ 0 = \tilde{X}(j\omega, n-1) - a\tilde{X}(j\omega, n) \\ &= j\omega \tilde{X}(j\omega, n) \end{aligned}$$

$$\tilde{X}(j\omega, n-1) = (a+j\omega) \tilde{X}(j\omega, n)$$

$$\Rightarrow X(j\omega) = \tilde{X}(j\omega, n) = \frac{1}{(a+j\omega)^{n-1}} \tilde{X}(j\omega, 1)$$

$$\cancel{\tilde{X}} \quad e^{-at} u(t) \rightsquigarrow \tilde{X}(j\omega, 1) = \frac{1}{j\omega + a} \Rightarrow X(j\omega) = \frac{1}{(a+j\omega)^n}$$